

Lecture Scribe Min-Cut Algorithm



L10_S1.pdf

PDF

Make a lecture scribe for the ppt uploaded for exam preparation.

Limit yourself to given lecture slides only. Do not apply any external knowledge and do not introduce any part that is not clearly mentioned in the lecture slides.

As you write the lecture scribe, give step by step reasoning of definitions and proofs of derivations and algorithms or a worked example where it is written in the lecture. This step by step argument can be provided in the final scribe so that the content to be covered in the lectures can be easily examined again. All the reasoning must be made solely out of the provided lecture slides.

Include exactly what is taught in the lecture, such as:

Definitions and notation

Assumptions and conditions

The statement of theorems or the results.

Written step by step proofs or proof sketches.

Proper explanation of worked out examples.

If the lecture includes algorithms, reproduce them exactly as they appear in the slides

Do not include any intuition, explanations, examples, and interpretations that are absent in the lecture. Do not make simple or paraphrase technical material.

Arrange the scribe according to the lecture in order of clarity with clear headings to enable easy use in the actual revision.

Use formal tone in the academics and have the scribe precise and reflective of the lecture.

Prepare the lecture scribe in such a way that the student who is merely reading this paper can be sure that she/he is repeating what was really taught in the classroom.

CSE400 – Fundamentals of Probability in Computing

Lecture 10: Randomized Min-Cut Algorithm

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Date: February 5, 2026

1. Outline of the Lecture

The lecture covers the following topics in order:

- 1. Min-Cut Problem**
- 2. Why use min-cut?**
- 3. What is min-cut?**

4. Successful min-cut run
5. Unsuccessful min-cut run
6. Max-Flow Min-Cut Theorem
7. Deterministic Min-Cut Algorithm
8. Stoer–Wagner Minimum Cut Algorithm
9. Pseudocode
10. Randomized Min-Cut Algorithm
11. Why randomized algorithm?
12. Karger's Randomized Algorithm
13. Pseudocode
14. Comparison: Deterministic Min-Cut vs Randomized Min-Cut Algorithm
15. Theorem for min-cut set
16. Python Simulation
17. Class Participation

The scribe follows this exact sequence.

2. Min-Cut Problem

2.1 Why Use Min-Cut?

The lecture states:

We use the min-cut algorithm in various applications to solve problems related to network connectivity, reliability, and optimization.

This establishes the purpose of the min-cut algorithm:

- It is applied in **network connectivity**.
- It is applied in **network reliability**.
- It is applied in **optimization problems**.

The lecture then specifies applications:

(1) Network Design

Min-cut helps in improving the efficiency of communication and optimizing network flow.

The algorithm is used in network design to find the minimum capacity cut.

Thus:

- Objective: Improve communication efficiency.
- Objective: Optimize network flow.
- Method: Identify the minimum capacity cut.

(2) Communication Networks

For understanding the vulnerability of the networks to failures minimum cut can be useful.

It helps building robust and fault-tolerant communication networks.

Thus:

- It evaluates network vulnerability.
- It assists in constructing robust systems.
- It supports fault tolerance.

(3) VLSI Design

In Very Large Scale Integration (VLSI) design, the algorithm is useful for partitioning circuits into smaller components leading to reduced interconnectivity complexity.

Thus:

- Used in circuit partitioning.
- Leads to reduced interconnectivity complexity.

Reference in lecture:

Section 1.5, Application: A Randomized Min-Cut Algorithm in *Probability and Computing (2nd Edition)*.

3. What is Min-Cut?

3.1 Definition: Cut-Set

A cut-set in a graph is:

A set of edges whose removal breaks the graph into two or more connected components.

Thus:

- Input: Graph.
- Operation: Remove a set of edges.
- Result: Graph splits into ≥ 2 connected components.

This defines the structural property of a cut.

3.2 Definition: Minimum Cut

Given:

$$G = (V, E)$$

where:

- V = set of vertices
- E = set of edges
- n = number of vertices

The **minimum cut (min-cut) problem** is:

| To find a minimum cardinality cut-set in G .

Thus:

- Among all cut-sets,
 - Select the one with smallest number of edges,
 - That is the min-cut.
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3.3 Sensitivity of Randomized Algorithms

The lecture states:

Min-cut algorithms, such as Karger's algorithm, are random and can be sensitive to the initial choice of edges. If the algorithm happens to contract critical edges early in the process, it might find a smaller cut.

Thus:

- Randomness affects edge selection.
- Early contraction of critical edges can change outcome.

No additional interpretation is introduced beyond what is stated.

4. Edge Contraction

The lecture defines the main operation.

4.1 Definition: Edge Contraction

The main operation in the algorithm is **edge contraction**.

Edge contraction is:

An operation that removes an edge from a graph while simultaneously merging the two vertices.

Let the edge be:

$$(u, v)$$

Then contraction consists of:

1. Merge vertices u and v into one vertex.
2. Eliminate all edges connecting u and v .
3. Retain all other edges.

After contraction:

- The new graph may have **parallel edges**.
- The new graph has **no self-loops**.

These properties are explicitly stated.

5. Successful Min-Cut Run

Definition from lecture:

A successful min cut run refers to the success in the outcome of an algorithm designed to find the minimum cut in a graph.

Thus:

- Algorithm runs.
- Correctly identifies the minimum cut.
- This run is called successful.

The slide includes a figure titled:

Figure: Successful Run

No further reasoning is added beyond the definition given.

6. Unsuccessful Min-Cut Run

Definition from lecture:

An unsuccessful min-cut run refers to an iteration of a min-cut algorithm where the algorithm fails to correctly identify the minimum cut of a given graph.

Thus:

- Algorithm performs a run.
- It does not correctly identify the minimum cut.
- That iteration is unsuccessful.

The slide includes a figure titled:

Figure: Unsuccessful Run

7. Max-Flow Min-Cut Theorem

7.1 Statement of Theorem

The lecture states:

In a flow network, the maximum amount of flow passing from the source to the sink is equal to the total weight of the edges in a minimum cut.

Thus:

$$\text{Maximum Flow} = \text{Minimum Cut Capacity}$$

7.2 Definitions in the Theorem

Capacity of a cut

Capacity of a cut =

The sum of the capacity of the edges in the cut that are oriented from a vertex $\in X$ to a vertex $\in Y$.

Thus:

- Partition vertices into sets X and Y .
 - Consider edges from $X \rightarrow Y$.
 - Sum their capacities.
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Minimum cut

Minimum cut =

The cut in the network that has the smallest possible capacity.

Minimum cut capacity

Minimum cut capacity =

The capacity of the minimum cut.

Maximum flow

Maximum flow =

The largest possible flow from source S to sink T .

8. Deterministic Min-Cut Algorithm

8.1 Stoer–Wagner Min-Cut Algorithm

Let:

- s and t be two vertices of graph G .
- $G/\{s, t\}$ be the graph obtained by merging s and t .

The lecture states:

A minimum cut of G can be obtained by taking the smaller of:

1. A minimum $s - t$ -cut of G , and
 2. A minimum cut of $G/\{s, t\}$.
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Reasoning given in lecture

Case 1:

There exists a minimum cut of G that separates s and t .

Then a minimum $s - t$ -cut of G is a minimum cut of G .

Case 2:

There is no minimum cut separating s and t .

Then a minimum cut of $G/\{s, t\}$ gives the result.

This completes the theorem explanation exactly as presented.

9. Pseudocode (Deterministic)

Algorithm 1: MinimumCutPhase(G, a)

1. $A \leftarrow \{a\}$
 2. While $A \neq V$:
 - Add to A the most tightly connected vertex
 3. Return the cut weight as the “cut of the phase”
-

Algorithm 2: MinimumCut(G)

1. While $|V| \geq 1$:
 - Choose any a from V
 - MinimumCutPhase(G, a)
 - If the cut-of-the-phase is lighter than the current minimum cut:
 - Store the cut-of-the-phase as the current minimum cut
 - Shrink G by merging the two vertices added last
 2. Return the minimum cut
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10. Randomized Min-Cut Algorithm

10.1 Why Randomized Algorithm?

The lecture states:

Randomized algorithms provide:

- A probabilistic guarantee of success.
- A more accurate estimate of the minimum cut with fewer iterations.

Additional stated properties:

- Efficiency
- Parallelization
- Approximation Guarantees
- Avoidance of Worst-Case Instances
- Heuristic Nature
- Robustness

No additional elaboration is introduced.

11. Karger's Randomized Algorithm

The lecture introduces:

Karger's Randomized Algorithm

(Title slide provided; no extra derivation included in slide text.)

12. Comparison: Deterministic vs Randomized

The lecture states:

Any specific problem is responsible for deciding the approach.

Deterministic Min Cut

- Always guarantees an exact minimum cut.
- May have higher time complexity for large graphs.
- Stoer-Wagner time complexity:

$$O(V \cdot E + V^2 \log V)$$

where:

- V = number of vertices
 - E = number of edges
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Randomized Min Cut

- Provides an approximate minimum cut with high probability.
- Karger's algorithm time complexity:

$$O(V^2)$$

where:

- V = number of vertices

13. Theorem for Min-Cut Set

The lecture states:

The algorithm outputs a min-cut set with probability at least:

$$\frac{2}{n(n-1)}$$

This is the probability bound given in the slide.

14. Python Simulation

Instructions from lecture:

- Open Campuswire Post regarding today's lecture (L10).
 - Download the `.ipynb` file pasted in the comments.
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15. Conclusion

This lecture covered:

- Definition of cut-set and min-cut.
- Edge contraction operation.
- Successful and unsuccessful min-cut runs.
- Max-Flow Min-Cut Theorem.

- Deterministic approach (Stoer–Wagner).
 - Pseudocode.
 - Randomized approach (Karger's algorithm).
 - Complexity comparison.
 - Probability theorem for randomized min-cut.
 - Python simulation instructions.
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End of Lecture 10 Scribe

Prepared strictly from the provided lecture slides